



THE MANY USES OF Apache Cassandra

This article covers Apache Cassandra's data model and tells you when to choose it as a NoSQL database for an application. Developers using NoSQL as a database or those looking for a suitable NoSQL database for their applications will find this informative.

Apache Cassandra is an open source, distributed, massively scalable NoSQL database. It is designed to handle large volumes of structured, semi-structured and unstructured data across multiple data centres, and it supports the cloud. Cassandra offers capabilities like continuous availability, linear scalability and operational simplicity across many commodity servers with no single point of failure. Its powerful, dynamic data model is designed for maximum flexibility and fast response times.

History

Apache Cassandra was originally developed by Avinash Lakshman (one of the authors of Amazon's Dynamo) and Prashant Malik at Facebook for inbox search. Cassandra was published as an open source project on Google Code in July 2008. It was accepted into the Apache Incubator in March 2009 and, since February 2010, has been an *Apache top-level project*.

Features

Massively scalable architecture: Cassandra has a masterless design where all nodes are the same. This provides operational simplicity and easy scale-out. There is no single point of failure.

Multiple data centre replication: Cassandra gives great support for replication across multiple data centres (in multiple geographies) and multi-cloud availability zones for writes/reads.

Linear scale performance: Users can increase performance by adding new nodes without any downtime or interruption to applications. For example, if one node can handle 50,000 transactions per second, two nodes will support 100,000 transactions/second and four nodes will tackle 200,000 transactions/second, and so on.

Tunable data consistency: Cassandra supports configured consistency levels to manage availability versus data accuracy. We can configure consistency on a Cassandra cluster, data centre, or per individual read or write operation.

Tunable consistency is one of the strongest features of Cassandra. There are two types of consistency levels – strong or eventual. To ensure that data is written and read correctly, Cassandra extends the concept of eventual consistency by offering tunable consistency. Tunable data consistency allows individual read or write operations to be as strongly consistent as required by the client application. The consistency level of each read or write operation can be set, so that the data returned is more or less consistent, based on need.

Data compression: Data is compressed up to 80 per cent without any performance overhead.

Cassandra Query Language: This is an SQL-like language that allows the running of SQL-like queries in traditional databases.

When should Cassandra be used?

Apache Cassandra can be used for various applications. Here are some use cases where it would be the best choice over other NoSQL databases.